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ECHS FRAUD ANALYTICS  
Hospital Specialty Misuse & Service Empanelment Analysis  
MODULE 12 COMPREHENSIVE REPORT — FY 2021–2026 EDITION  
Classification  
RESTRICTED  
Period  
FY 2021–26  
Checks Run  
5 Checks  
Entities Flagged  
{fmt(total\_hospitals\_flagged)}  
IIT KANPUR — Data Analytics & Fraud Intelligence Division  
{today\_str} | Ex-Servicemen Contributory Health Scheme (ECHS)  
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## Executive Summary

This report investigates hospital specialty misuse — cases where facilities bill for services outside their designated category, and anomalous deduction patterns across hospitals. Five primary fraud patterns are confirmed.

Hospitals Flagged  
{fmt(total\_hospitals\_flagged)}  
across 5 fraud checks  
Total Exposure  
{cr(total\_exposure)}  
estimated financial risk  
Checks w/ Findings  
{patterns\_found}  
of 5 checks  
Analysis Period  
5 Yrs  
FY 2021–2026

## Five Fraud Checks — Summary

| {th("#","Check","What It Detects","Cases Flagged","Risk")} |  |                     |                        |
|------------------------------------------------------------|--|---------------------|------------------------|
|                                                            |  | Type-3 facilities   |                        |
| Specialty                                                  |  | (dental clinics and |                        |
| 1 Category Billing                                         |  | diagnostic labs)    | {fmt(len(p01a))}       |
| Fraud                                                      |  | facilities          | {risk_txt("CRITICAL")} |

|                                |                                                                                               |                                      |                                 |
|--------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------|---------------------------------|
|                                | billing IPD admissions.                                                                       |                                      |                                 |
| NABH vs Non-NABH Divergence    | NABH-accredited facilities show consistently lower deduction rates, but some anomalies exist. | <b>{{fmt(len(p02b))}}</b> anomalous  | <b>{{risk_txt("HIGH")}}</b>     |
| 3 Out of Scope Empanelment     | Hospitals billing for IPD services when they are NOT empaneled for IPD.                       | <b>{{fmt(len(p03b))}}</b> facilities | <b>{{risk_txt("CRITICAL")}}</b> |
| 4 Year-over-Year Billing Spike | Sudden 100%+ annual growth in claim amount indicating deliberate strategy change.             | <b>{{fmt(len(p04a))}}</b> facilities | <b>{{risk_txt("HIGH")}}</b>     |
| High-Value Claims at Low-Tier  | Type-3 facilities billing abnormally high individual claim amounts (over ₹1L).                | <b>{{fmt(len(p05b))}}</b> facilities | <b>{{risk_txt("HIGH")}}</b>     |

## Immediate Recommended Actions

1. **Audit all Type-3 IPD Billings** (**{{fmt(len(p01a))}}** cases) — Deny all IPD claims from dental clinics unless explicitly authorized.
2. **Investigate anomalous NABH hospitals** (**{{fmt(len(p02b))}}** cases) — A high deduction rate despite NABH status suggests potential bill padding.
3. **Block out-of-scope empanelment billing** (**{{fmt(len(p03b))}}** facilities) — Enforce system-level blocks preventing non-IPD hospitals from submitting IPD bills.
4. **Review YoY billing spike** (**{{fmt(len(p04a))}}** facilities) — Request clinical justification for hospital billing that has doubled year-over-year.
5. **Flag high-value Type-3 claims** (**{{fmt(len(p05b))}}** facilities) — Require manual pre-authorization for any Type-3 claim exceeding ₹50k.

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CHECK 1

Type-3 Facilities (Dental/Diag)

Specialty Category Billing Fraud (IPD)

**Description:** Type-3 facilities (dental clinics and diagnostic laboratories) are billing inpatient (IPD) admissions — a service category they are not licensed or

equipped to provide. These facilities should only handle outpatient procedures. **All such IPD claims represent severe policy violations.**

Table 1.1 — Type-3 Facilities Billing IPD Claims (Top 15 by Claimed Amount)

 {th("Hospital Name","Type","City","State","IPD Claims","Claimed Amt","Ded. %")} |

\\n\\n\\n\\n\\n\\n\\n\\n\\n\\n\\n\\n\\n\\n\\n\\n\\n\\n\\n\\n

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc',''))} {safe(r.get('city',''))} {safe(r.get
```

```
[ :30])}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc',''))} {safe(r.get('city',''))} {safe(r.get
```

```
[ :30] }
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc',''))} {safe(r.get('city',''))} {safe(r.get
```

```
[ :30]]
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc',''))} {safe(r.get('city',''))} {safe(r.get
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```
[ :30] }
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc',''))} {safe(r.get('city',''))} {safe(r.get
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[ :30] }
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc',''))} {safe(r.get('city',''))} {safe(r.get
```

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[ :30]}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc',''))} {safe(r.get('city',''))} {safe(r.get
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[ :30] }
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc',''))} {safe(r.get('city',''))} {safe(r.get
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```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc',''))} {safe(r.get('city',''))} {safe(r.get('state',''))} [:30]}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc',''))} {safe(r.get('city',''))} {safe(r.get('state',''))}
[:30]}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc',''))} {safe(r.get('city',''))} {safe(r.get('state',''))}
[:30]}
```

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{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc',''))} {safe(r.get('city',''))} {safe(r.get('state',''))}
[:30]}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc',''))} {safe(r.get('city',''))} {safe(r.get('state',''))}
[:30]}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc',''))} {safe(r.get('city',''))} {safe(r.get('state',''))}
[:30]}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc',''))} {safe(r.get('city',''))} {safe(r.get('state',''))}
```

## Key Findings

**IPD Claims at Dental Clinics:** Dental clinics and diagnostic centers are solely empaneled for OPD (Outpatient Department) procedures. Billing for IPD admissions from a Type-3 facility implies fraudulent coding or systemic misuse of the hospital classification.

Facilities flagged here must be immediately audited to ensure they are not admitting patients overnight in unlicensed premises.

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## CHECK 2

## NABH Divergence Anomaly

(>15% Deduction in NABH)

## NABH Accreditation & High Deduction Anomalies

**Description:** NABH-accredited facilities are expected to adhere to higher billing standards, typically resulting in lower deduction rates (the difference between claimed and approved amounts). When an NABH hospital consistently suffers extremely high deductions (over 15%), it indicates chronic over-billing, unbundled charging, or billing for non-entitled items despite their accreditation.

Table 2.1 — NABH Hospitals with Anomalously High Deduction Rates (Top 15)

| Hospital Name | Type | City | Claims | Claimed (Lakh) | Approved (Lakh) | Ded. % |
|---------------|------|------|--------|----------------|-----------------|--------|
| 1             | 2    | 3    | 4      | 5              | 6               | 7      |
| 8             | 9    | 10   | 11     | 12             | 13              | 14     |
| 15            | 16   | 17   | 18     | 19             | 20              | 21     |
| 22            | 23   | 24   | 25     | 26             | 27              | 28     |
| 29            | 30   | 31   | 32     | 33             | 34              | 35     |
| 36            | 37   | 38   | 39     | 40             | 41              | 42     |
| 43            | 44   | 45   | 46     | 47             | 48              | 49     |
| 50            | 51   | 52   | 53     | 54             | 55              | 56     |
| 57            | 58   | 59   | 60     | 61             | 62              | 63     |
| 64            | 65   | 66   | 67     | 68             | 69              | 70     |
| 71            | 72   | 73   | 74     | 75             | 76              | 77     |
| 78            | 79   | 80   | 81     | 82             | 83              | 84     |
| 85            | 86   | 87   | 88     | 89             | 90              | 91     |
| 92            | 93   | 94   | 95     | 96             | 97              | 98     |
| 99            | 100  | 101  | 102    | 103            | 104             | 105    |

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get(
```

```
[ :30]) [ :20]) }
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get(
```

```
[ :30]) [ :20]) }
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get(
```

```
[ :30]} [ :20]}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get(
```

```
[ :30]} [ :20]}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get(
```

```
[ :30]} [ :20]}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get(
```

```
[ :30]} [ :20]}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get('
[:30])} [:20])}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get(
```

```
[ :30]} [ :20]}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get(
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```
[ :30]} [ :20]}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get(
```

```
[·30]} [·20]}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get(
```

```
[·30]} [·20]}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get(
```

```
[·30]} [·20]}
```

```
{safe(r.get('city',''))} {safe(r.get(
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc',''))
[:30]} [:20]}
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get(
[:30]} [:20]}
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get(
[:30]} [:20]}
```

## Key Findings

**Ded. % (Deduction Percentage):** A high deduction rate in an NABH hospital contradicts the expected billing quality of an accredited facility. These hospitals may be utilizing their NABH status to attract ECHS patients while systematically inflating bills to maximize approved payouts.

Military Hospitals (Type M) often show very high deductions, but they are not the focus here; these are private empanelled NABH facilities exhibiting predatory billing behaviors.

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### CHECK 3

## Services vs Claims

(IPD Billing by Non-IPD Facilities)

### Services Outside Empaneled Scope

**Description:** Hospitals must be explicitly empaneled for the services they provide. This check identifies hospitals that are billing for inpatient (IPD) services but are **NOT** empaneled for IPD in the ECHS registry. This represents direct scope creep fraud.

Table 3.1 — IPD Claims without IPD Empanelment (Top 15)

 {th("Hospital","Type","City","IPD Claims","Claimed Amt","Empaneled Services")} |

\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n

|                                               |                                                |                                       |                                       |
|-----------------------------------------------|------------------------------------------------|---------------------------------------|---------------------------------------|
| <code>{safe(r.get('hospital_name',''))</code> | <code>{safe(r.get('hosp_type_desc',''))</code> | <code>{safe(r.get('city',''))}</code> | <code>{safe(r.get('city',''))}</code> |
| <code>[:25]}</code>                           | <code>[:15]}</code>                            |                                       | <code>{safe(r.get('city',''))}</code> |
| <code>{safe(r.get('hospital_name',''))</code> | <code>{safe(r.get('hosp_type_desc',''))</code> | <code>{safe(r.get('city',''))}</code> | <code>{safe(r.get('city',''))}</code> |
| <code>[:25]}</code>                           | <code>[:15]}</code>                            |                                       | <code>{safe(r.get('city',''))}</code> |
| <code>{safe(r.get('hospital_name',''))</code> | <code>{safe(r.get('hosp_type_desc',''))</code> | <code>{safe(r.get('city',''))}</code> | <code>{safe(r.get('city',''))}</code> |
| <code>[:25]}</code>                           | <code>[:15]}</code>                            |                                       | <code>{safe(r.get('city',''))}</code> |
| <code>{safe(r.get('hospital_name',''))</code> | <code>{safe(r.get('hosp_type_desc',''))</code> | <code>{safe(r.get('city',''))}</code> | <code>{safe(r.get('city',''))}</code> |
| <code>[:25]}</code>                           | <code>[:15]}</code>                            |                                       | <code>{safe(r.get('city',''))}</code> |
| <code>{safe(r.get('hospital_name',''))</code> | <code>{safe(r.get('hosp_type_desc',''))</code> | <code>{safe(r.get('city',''))}</code> | <code>{safe(r.get('city',''))}</code> |
| <code>[:25]}</code>                           | <code>[:15]}</code>                            |                                       | <code>{safe(r.get('city',''))}</code> |
| <code>{safe(r.get('hospital_name',''))</code> | <code>{safe(r.get('hosp_type_desc',''))</code> | <code>{safe(r.get('city',''))}</code> | <code>{safe(r.get('city',''))}</code> |
| <code>[:25]}</code>                           | <code>[:15]}</code>                            |                                       | <code>{safe(r.get('city',''))}</code> |

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc',''))  
[:25]} [:15]]  
  
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get('state',''))}  
[:25]} [:15]]  
  
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get('state',''))}  
[:25]} [:15]]  
  
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get('state',''))}  
[:25]} [:15]]  
  
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get('state',''))}  
[:25]} [:15]]  
  
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get('state',''))}  
[:25]} [:15]]
```

## Key Findings

**Empaneled Services:** Indicates the specific medical facilities the hospital is legally permitted to bill ECHS for. If "IPD" or "Indoor" is missing from this list, the hospital is strictly an OPD facility.

Billing IPD without IPD empanelment is a critical compliance failure. ECHS processing centers should implement hard-blocks to reject IPD claims from these facilities automatically.

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## CHECK 4

## YoY Financial Trend

(>100% Annual Amount Growth)

## Year-over-Year Billing Spike by Hospital

**Description:** Detects hospitals demonstrating a sudden **100%+ annual growth** in claim amounts without a corresponding growth in their registered patient base or empanelled capacity. A cliff-edge spike of this magnitude indicates a deliberate change in aggressive billing strategies rather than organic demographic growth.

Table 4.1 — Hospital YoY Billing Spike (Top 15)

```
{th("Hospital","City","Prev Yr Amt","Curr Yr Amt","Amt Growth %","Claims Growth %")) \n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('city',''))} {cr(float(r.get('prev_claimed_lakh', {cr(float(0))*100000)))*100000)}  
[:30]}  
  
{safe(r.get('hospital_name','')) {safe(r.get('city',''))} {cr(float(r.get('prev_claimed_lakh', {cr(float(0))*100000)))*100000)}  
[:30]}
```

```
{safe(r.get('hospital_name',''))} {safe(r.get('city',''))} {cr(float(r.get('prev_claimed_lakh', '{cr(float(
```

```
{safe(r.get('hospital_name',''))} {safe(r.get('city',''))} {cr(float(r.get('prev_claimed_lakh',{cr(float(0))*100000})))*10
```

```
{safe(r.get('hospital_name',''))} {safe(r.get('city',''))} {cr(float(r.get('prev_claimed_lakh',{cr(float(0))*100000})))*10
```

```
{safe(r.get('hospital_name',''))} {safe(r.get('city',''))} {cr(float(r.get('prev_claimed_lakh',{cr(float(0))*100000})))*10
```

```
{safe(r.get('hospital_name',''))} {safe(r.get('city',''))} {cr(float(r.get('prev_claimed_lakh',{cr(float(0))*100000})))*10
```

```
{safe(r.get('hospital_name',''))} {safe(r.get('city',''))} {cr(float(r.get('prev_claimed_lakh',{cr(float(r.get('prev_claimed_lakh','0'))*100000)})))*10
```

```
{safe(r.get('hospital_name',''))} {safe(r.get('city',''))} {cr(float(r.get('prev_claimed_lakh',0))*100000)}
```

```
{safe(r.get('hospital_name',''))} {safe(r.get('city',''))} {cr(float(r.get('prev_claimed_lakh',0))*100000)}
```

```
{safe(r.get('hospital_name',''))[:30]} {safe(r.get('city',''))} {cr(float(r.get('prev_claimed_lakh',0))*100000)}
```

```
{safe(r.get('hospital_name',''))} {safe(r.get('city',''))} {cr(float(r.get('prev_claimed_lakh',0))*100000)}
```

```
{safe(r.get('hospital_name',''))} {safe(r.get('city',''))} {cr(float(r.get('prev_claimed_lakh', '{cr(float(
```

  
[:30])}) \* 100000))}

```
{safe(r.get('hospital_name',''))} {safe(r.get('city',''))} {cr(float(r.get('prev_claimed_lakh', '{cr(float(
```

```
{safe(r.get('hospital_name',''))[:30]} {safe(r.get('city',''))[:30]} {cr(float(r.get('prev_claimed_lakh',0))*100000)} {cr(float(r.get('prev_claimed_lakh',0))*100000)}
```

## Key Findings

**Amt Growth %:** Hospitals doubling their ECHS revenue in a single year must be audited. This often correlates with a new fraud ring operation, hiring of aggressive billing agents, or systematic upcoding of procedures.

Discrepancies where Amount Growth far exceeds Claims Growth indicate the hospital has started charging significantly more per patient (inflation of severity, unbundling).

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## CHECK 5

### Type-3 Facilities

(Extreme High-Value Claims >₹1L)

### High-Value Claims at Low-Tier Facilities

**Description:** Small or specialist-only hospitals (Type 3: Dental, Diagnostic, Polyclinics) billing individual claim amounts that exceed their clinical capabilities. A dental clinic billing ₹1 Lakh+ for a single admission is a strong indicator of fabricated bills or incorrect hospital classification.

Table 5.1 — Low-Tier Hospitals with High-Value Claims (Top 15)

```
{th("Hospital","Type","City","High Val Claims","Max Single Claim","Total Claimed")}
```

\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get('state',''))} }  
[:30]} [:15]}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get('state',''))}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get('state',''))}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get('state',''))}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get('state',''))} }[:30]} [:15]}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get('state',''))}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get('state',''))}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get('state',''))}
```

```
[ :30]} [ :15]}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get('state',''))}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get('state',''))}
```

```
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get('state',''))}
```



```

{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get('city',''))} {safe(r.get('city',''))}
[:30]} [:15]}
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get('city',''))} {safe(r.get('city',''))}
[:30]} [:15]}
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get('city',''))} {safe(r.get('city',''))}
[:30]} [:15]}
{safe(r.get('hospital_name','')) {safe(r.get('hosp_type_desc','')) {safe(r.get('city',''))} {safe(r.get('city',''))} {safe(r.get('city',''))}
[:30]} [:15]}

```

#### Key Findings

**Max Single Claim:** The maximum amount billed for a single patient episode. For Type-3 facilities, claims exceeding ₹50,000 are extremely rare and highly anomalous.

Facilities flagged here must provide detailed clinical justification for these high-value bills, as their infrastructure typically does not support procedures of this cost magnitude.

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